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College of Osteopathic Medicine

Osteopathic Considerations in Cardiology



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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Objectives

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- Be able to describe the autonomic processes of the cardiovascular system and how they affect the physiology of a patient and relate to viscerosomatic reflexes
- Be able to describe the lymphatic and musculoskeletal structures related to the cardiovascular system and how changes to these structures may affect the system.
- Be able to identify and describe the location of Chapman's points related to the cardiovascular system and how these can be used to diagnose and treat cardiovascular problems.
- The student will be able to describe the goals for treating a patient with osteopathic principles and be able to apply the 5 models of osteopathic treatment to treating patients with cardiovascular disease.

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Clinical Relevance:

You will be able to look for somatic dysfunctions in specific areas based on the patient's cardiac disease process and understand how and why to treat the patient's autonomic nervous system, lymphatic system and musculoskeletal system .

Models of Osteopathic Manipulation

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Biomechanical
(Postural/Musculoskeletal)

Behavioral
(Psychobehavioral)



Neurological

Metabolic-
Energy
(Bioenergy)

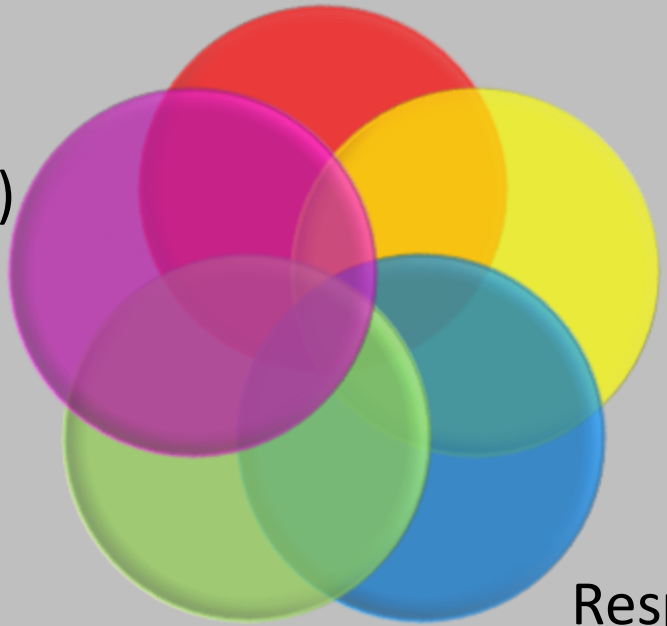
Respiratory/Circulatory

Neurological Model of Osteopathic Manipulation

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Biomechanical
(Postural/Musculoskeletal)

Behavioral
(Psychobeavioral)



Neurological

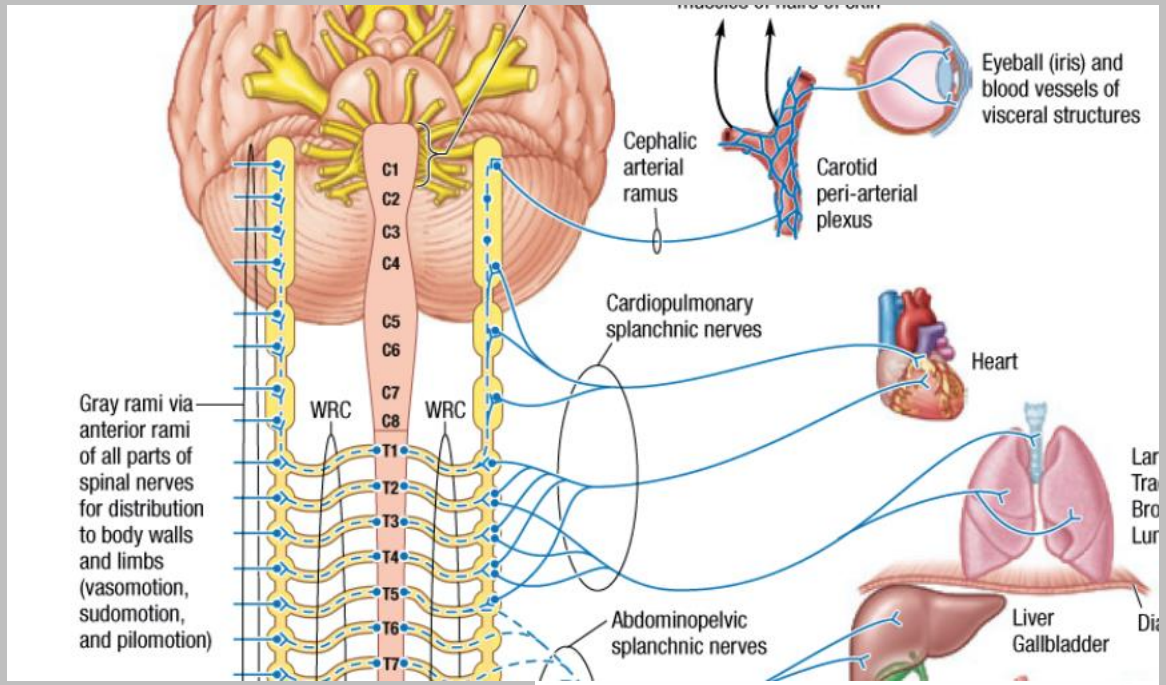
Metabolic-
Energy
(Bioenergy)

Respiratory/Circulatory

Sympathetic innervation to the heart

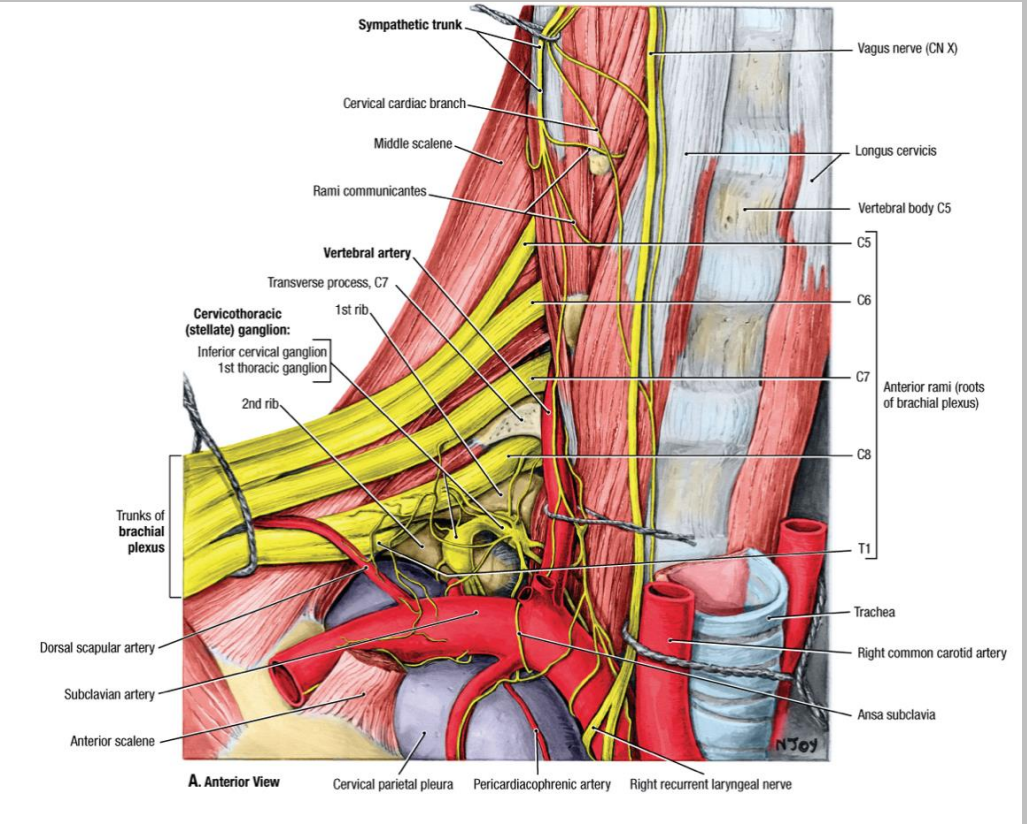
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T1-T5 synapsing with the thoracic and cervical chain ganglia.



Grant's Atlas of Anatomy, 14e, 2017
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- The cervicothoracic ganglia (Stellate ganglia) contributes sympathetics to the heart. Located anterior to TP of T1 and head of Rib 1 & sometimes Rib 2.



Grant's Atlas of Anatomy, 14e, 2017
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- Right sympathetic fibers → Right deep cardiac plexus → to innervate the right heart and SA node.
- Increased sympathetic tone to right sided heart → ?????

- Increased sympathetic tone to right sided heart → Predisposition for supra-ventricular tachyarrhythmias



- Left sympathetic fibers → Left deep cardiac plexus → AV node
- Increased sympathetic tone →????



- Increased sympathetic tone → predisposition for ectopic beats and ventricular fibrillation.
- Increased sympathetic tone to the coronary vessels has been linked to coronary vasospasm.

- Prolonged increased sympathetic tone to the heart decreases the heart's ability to form collateral circulation.

- An imbalance between the right and left stellate ganglionic sympathetic impulses to the heart has been postulated as being the mechanism for causing some ventricular arrhythmias in some long Q-T syndromes.

- Decreased sympathetic tone to the heart will lead to:
  - Decreased venous return

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Heart Parasympathetic innervation is via the Vagus N.

- Right Vagus n. innervates the SA node
- Left vagus innervates the AV node.

- What could increased parasympathetic tone to the right vagus (SA node) predispose the heart to?

- Increased parasympathetic tone to the right vagus → predisposes heart to bradyarhythmias



- Increased parasympathetic tone to the left vagus (AV node) predisposes the heart to ?????.

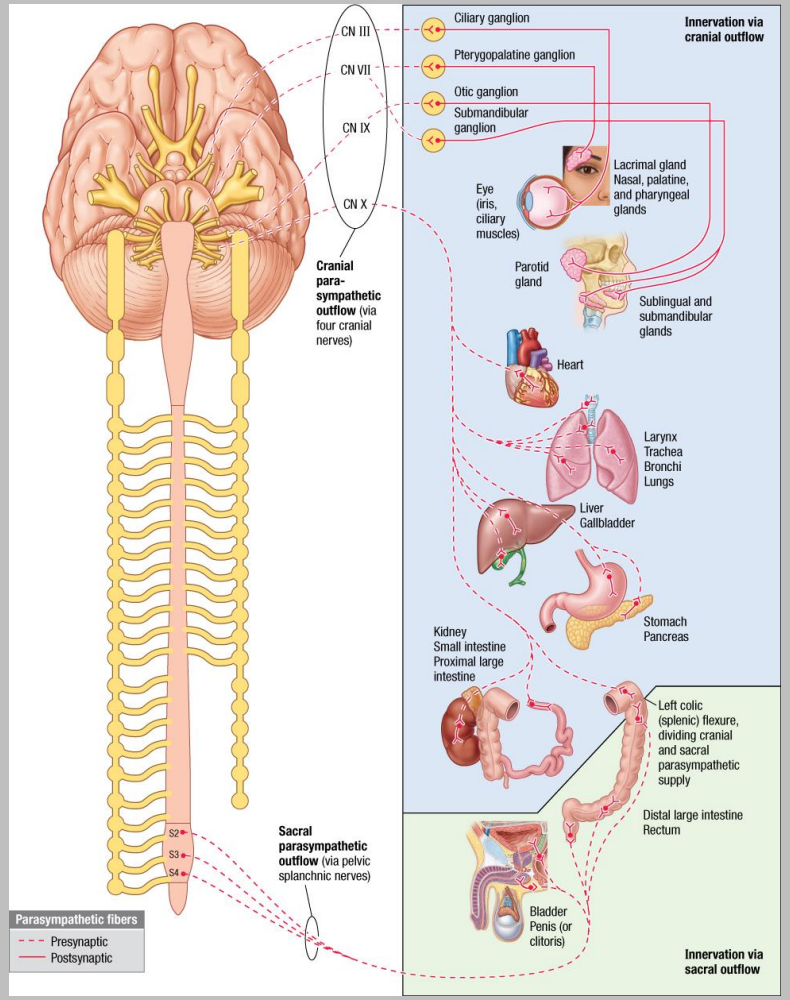
- Increased parasympathetic tone to the AV node predisposes the heart to AV block.



- Somatic dysfunctions at what areas could have an effect on the vagus nerve?

- Somatic dysfunctions at the OM suture, OA, AA and / or C2 can affect the vagal reflexes.
  - Jugular foramen (occiput & temporal bones)
  - C2 has a branch feeding the vagus

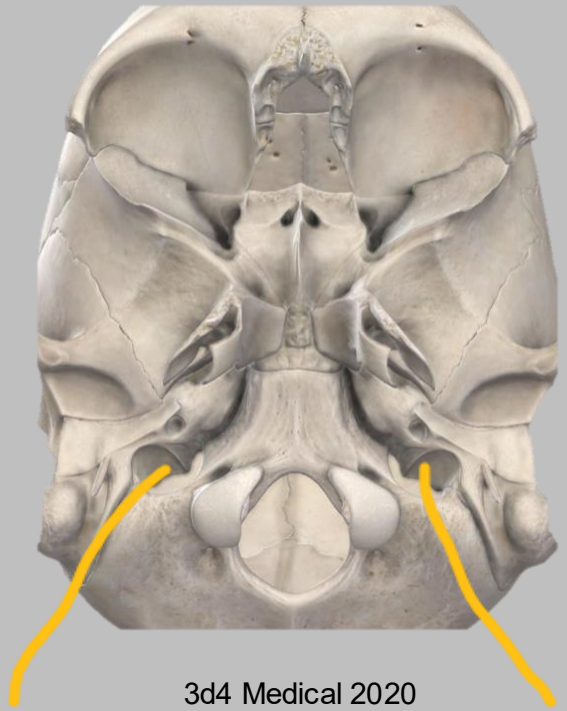
Distribution of Parasympathetic Nerve Fibers



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- Vagus exits the skull through the jugular foramen
- Basi-occiput & petrous portion of the temporal bone are borders of the jugular formamen
- Treatment of the O-A is an often-cited method for optimizing the Vagal Supply to the Viscera

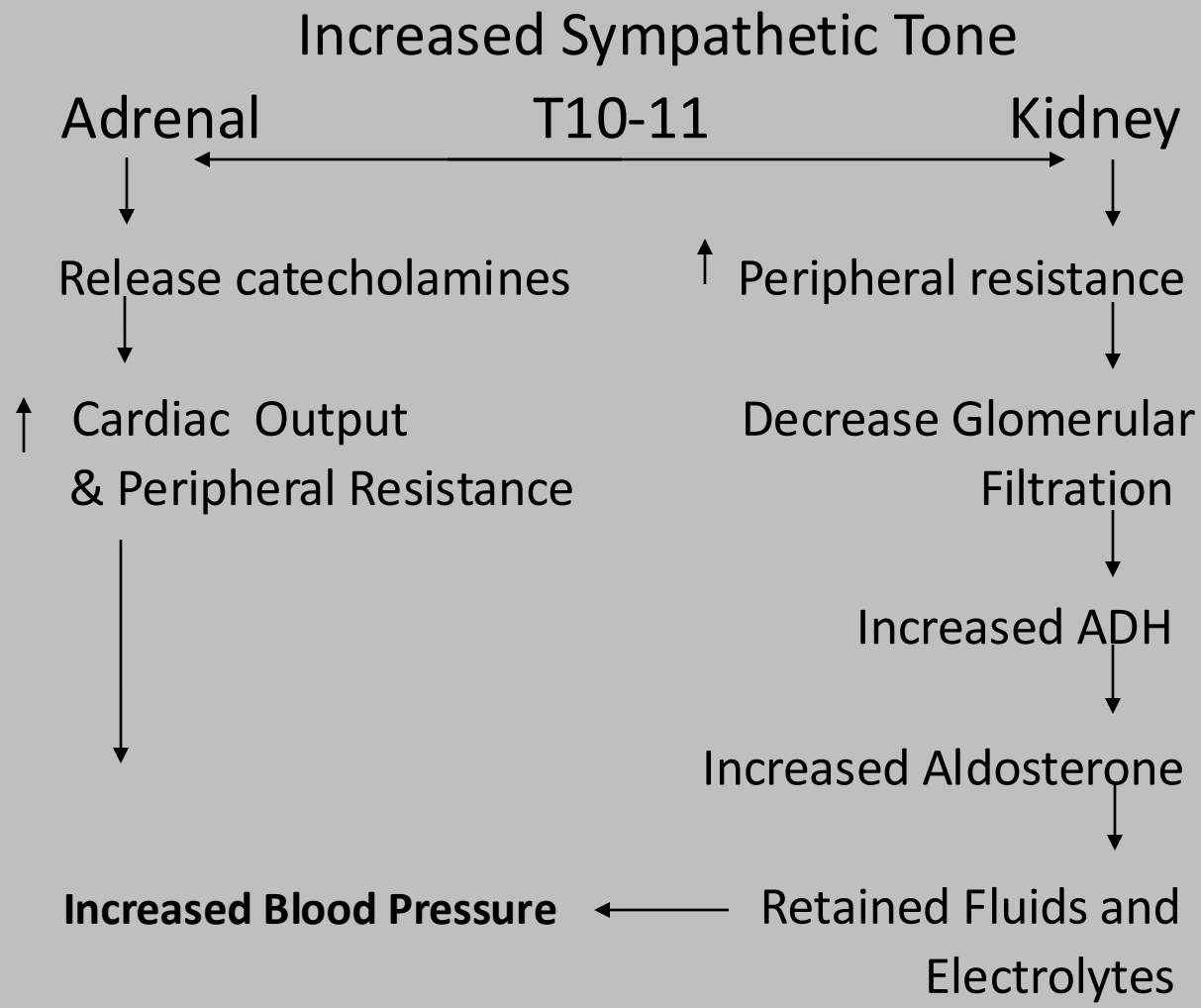


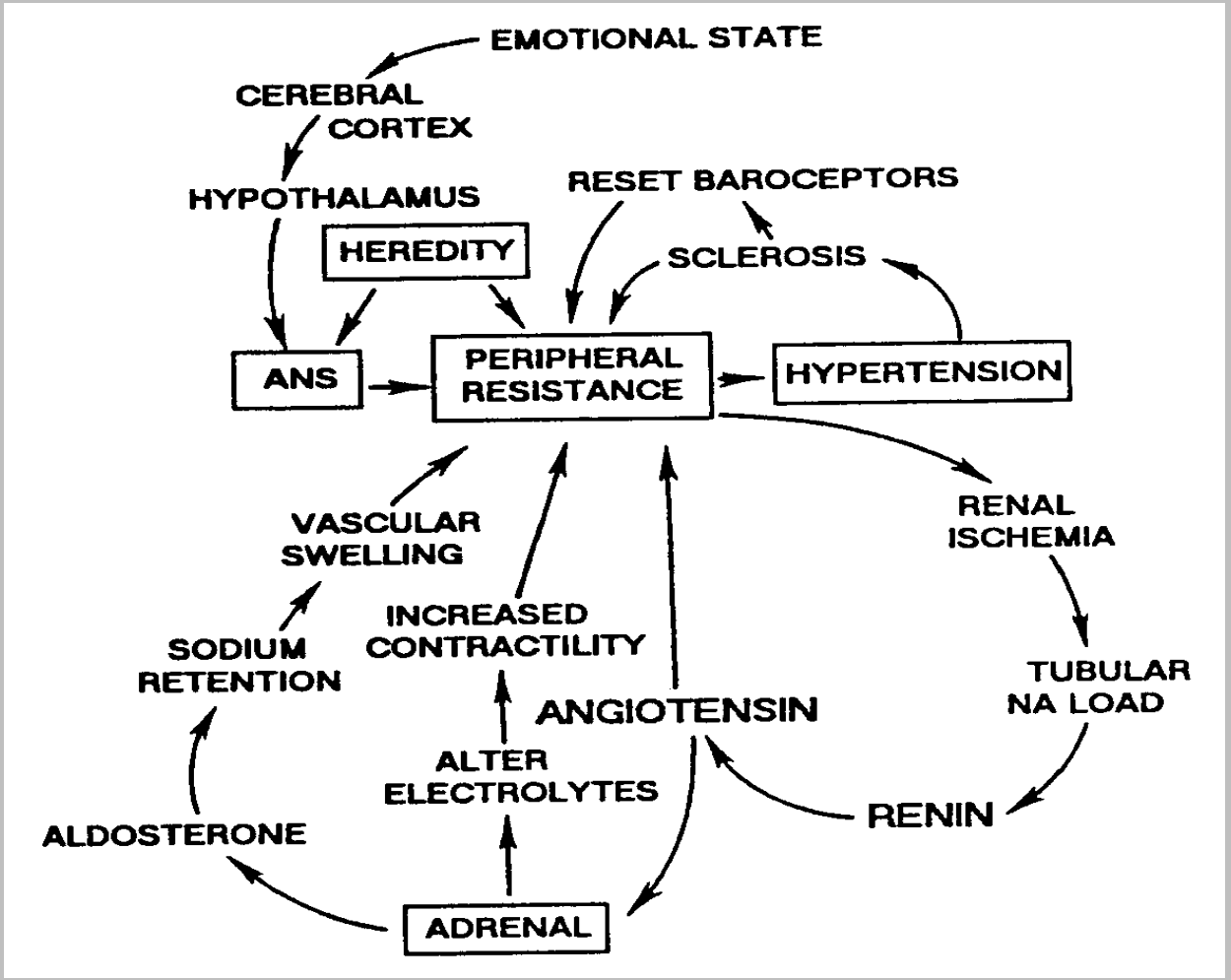
3d4 Medical 2020

Autonomics to the circulatory system

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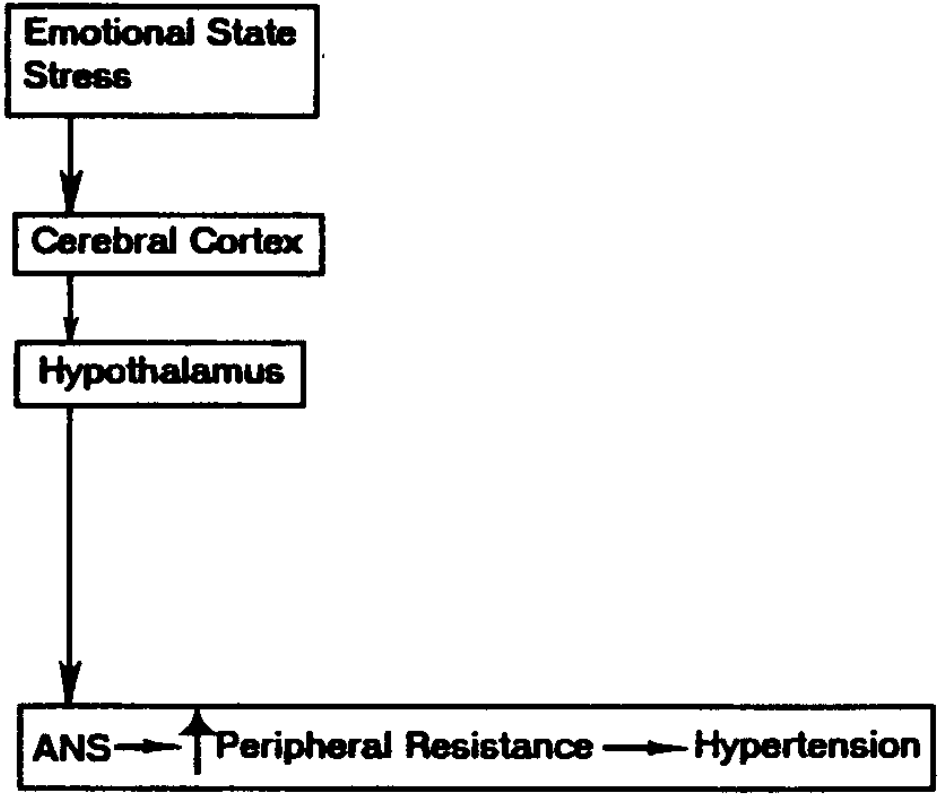
- Sympathetics to circulation are generally unopposed:
  - Except: submaxillary gland, parotid, blush region of the face, the tongue and the penis.
- Note: Increased sympathetics to T10-11 and its relationship to HTN.


Increased Sympathetic Tone
Adrenal
T10-11
Kidney
Release catecholamines
Peripheral resistance
Cardiac Output
& Peripheral Resistance
Decrease Glomerular
Filtration
Increased ADH
Increased Aldosterone
Increased Blood Pressure
Retained Fluids and
Electrolytes

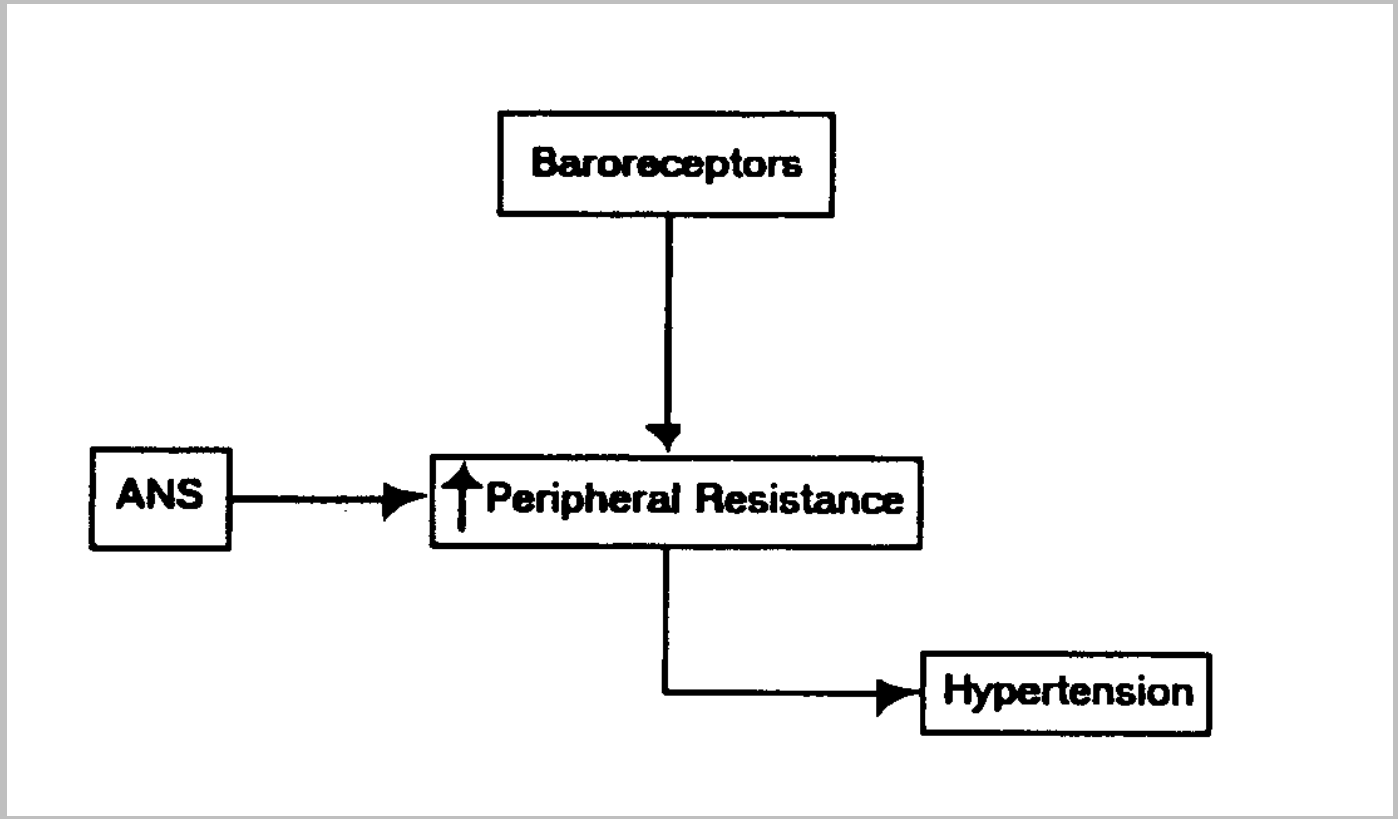


Kuchera. Osteopathic Considerations in Systemic Dysfunction. p. 62

NEUROGENIC HYPERTENSION



Kuchera. Osteopathic Considerations in Systemic Dysfunction. p. 62



Kuchera. Osteopathic Considerations in Systemic Dysfunction. p. 63

Respiratory/Circulatory Model

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Biomechanical
(Postural/Musculoskeletal)

Behavioral
(Psychbehavioral)



Neurological

Metabolic-
Energy
(Bioenergy)

Respiratory/Circulatory (includes lymphatics!)

Cardiovascular Lymphatics

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- The **endocardium, myocardium and epicardium** drain primarily through the right lymphatic duct
  - runs above apex of right lung. Drains into the right subclavian v.
- The **pericardium** drains primarily through the thoracic duct
  - runs above apex of left lung. Drains into the left subclavian v.

Cardiovascular Lymphatics

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- Thoracic duct runs from below diaphragm on right at the Cysterna Chyli and crosses to the left around the level of T5-T6.
  - Restriction here can cause decreased lymph drainage.
- Thoracic duct drains into the left subclavian v.

Cardiovascular Lymphatics (cont')

- The thoracic duct and large lymphatic vessels are under sympathetic control
  - Therefore: increased sympathetics can → **increase pumping of lymph. This effect is theorized to <u>be more to modify immune response rather than increase flow through vessels.</u>**

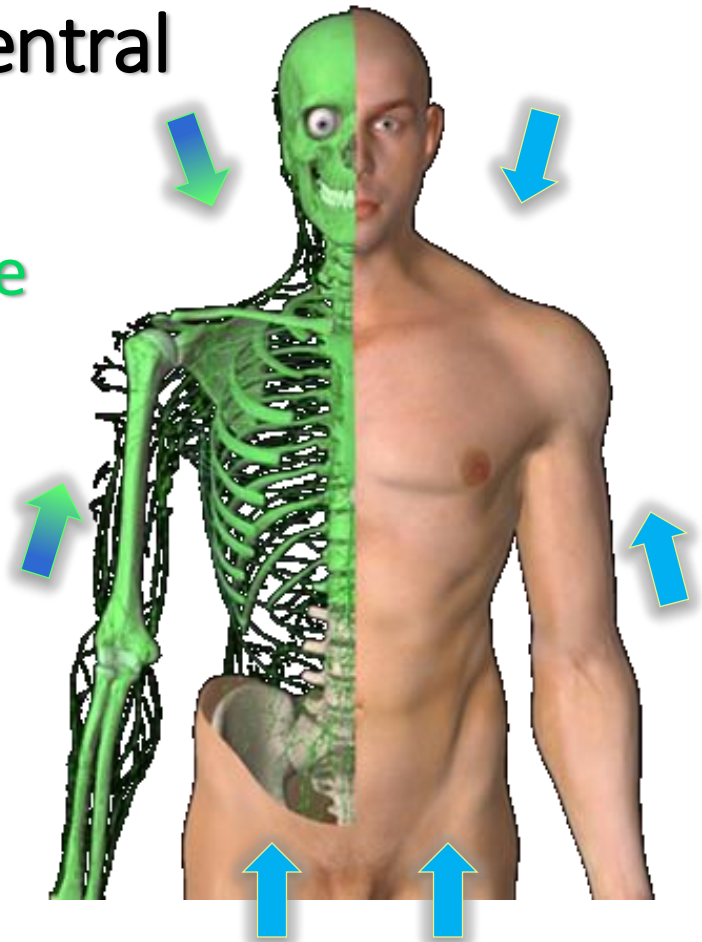
Lymph Drains from Peripheral Toward Central

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Heart, Lungs, Right upper ext, right face

–

Including endocardium, myocardium and epicardium



Rest of Body including pericardium

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- If the lymphaticovenous systems fail to function, the rate of edema of limbs and organs in congestive heart failure may be increased or resolution may be impeded.

- Impaired lymphatic drainage from the heart is known to severely compromise homeostasis and has been implicated in reduced collateral circulation and increased morbidity and mortality from ischemia and infection.

- Impaired flow of lymphatics can lead to:
  - Increased central venous pressure
  - Increased tissue congestion →decreased O_2 and nutrients to the myocardium
  - Increase the size from an MI
  - Increase the risk of subacute bacterial endocarditis

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- Study: “Active Drainage of Cardiac Lymph in Relation to Reduction in Size of Myocardial Infarction: - An Experimental Study”
  - Akira T, Uehara K, Fukuda S, Takenaka K, and Koga M
  - Angiology 1990 Dec; 41(12): 1029-36
  - Conclusions:
    - Hyaluronidase was administered to the canine heart with coronary artery ligature to remove the accumulated cardiac lymph from myocardium. The infarct risk area ratio was reduced significantly after the use of hyaluronidase and was accompanied by an increase in cardiac lymph flow. This experiment clarified the contribution of the cardiac lymph to the development of ischemic damage to the myocardium.

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- Study: “Active Drainage of Cardiac Lymph in Relation to Reduction in Size of Myocardial Infarction: - An Experimental Study”
  - Akira T, Uehara K, Fukuda S, Takenaka K, and Koga M
  - Angiology 1990 Dec; 41(12): 1029-36

(CONTINUED…)

Thoughts:

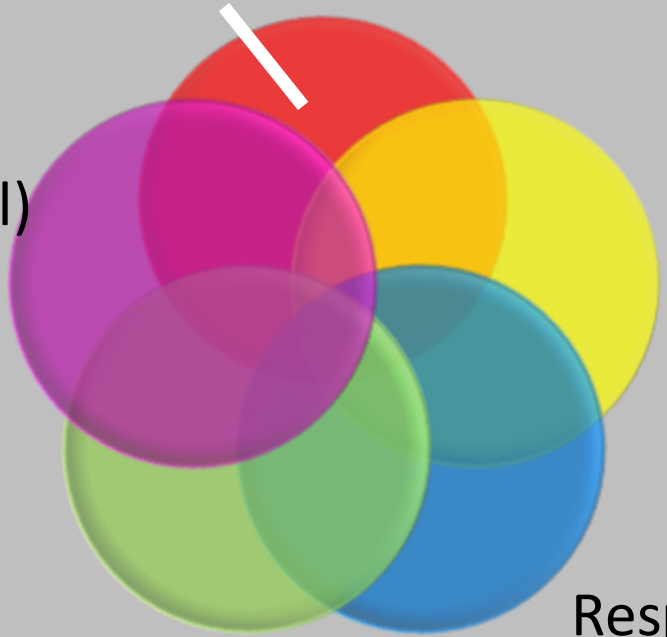
- Treating the body to maintain optimal lymphatic flow in a patient at risk for a myocardial infarction (MI) may help decrease the size of an MI if one does occur (research needed)
- Or, treating the lymphatics post-infarct may help decrease the area ultimately affected by the infarct. (research needed)

Biomechanical Model of Osteopathic Manipulation

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Biomechanical (Postural/Musculoskeletal)

Behavioral (Psychobehavioral)



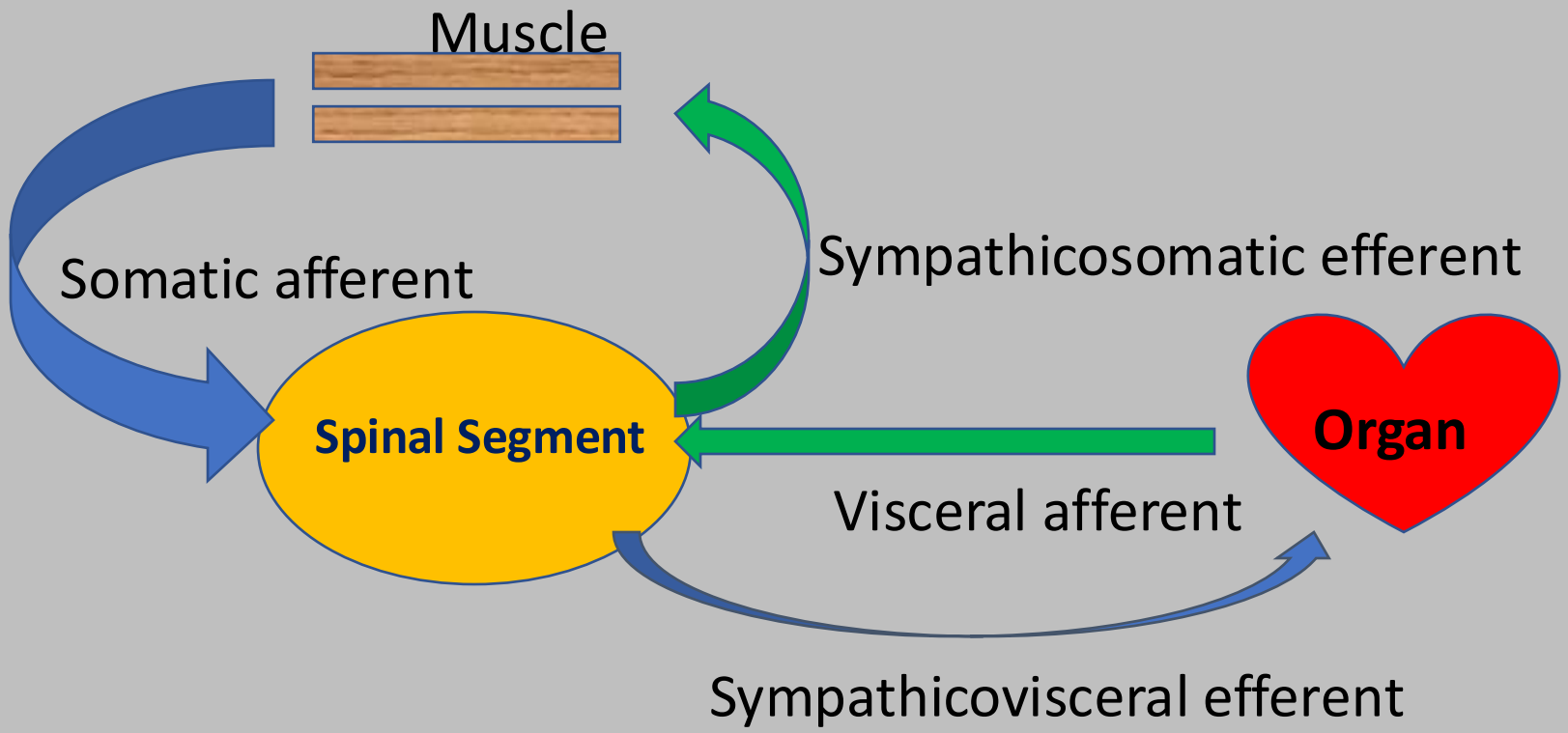
Neurological

Metabolic-Energy (Bioenergy)

Respiratory/Circulatory

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Viscerosomatic and Somatovisceral Reflexes



Musculoskeletal findings for Cardiac disease

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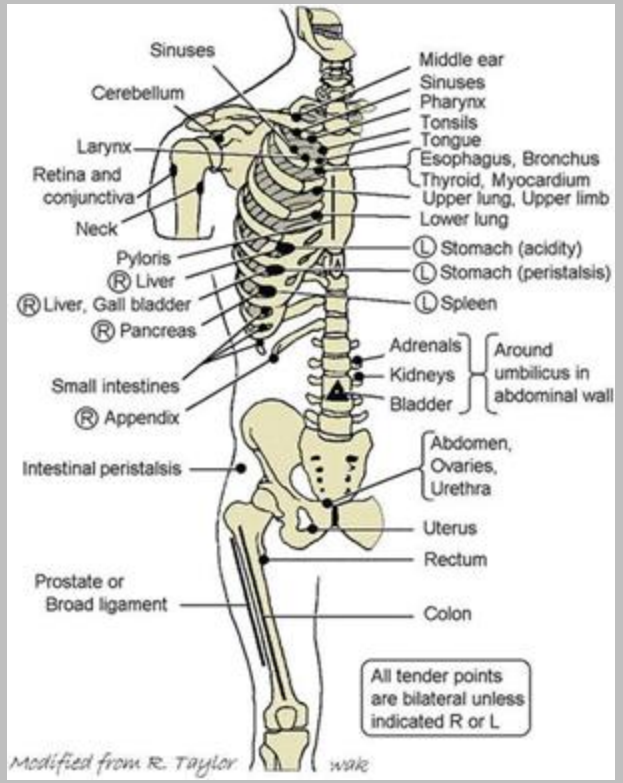
- Viscerosomatic reflexes at T1-5 on left (heart)
- Acute viscerosomatic reflexes
  - Warm, moist, boggy, edematous, + skin drag test, + erthyema test
- Chronic viscerosomatic reflexes
  - firm, dry sponginess, joint is “really stuck” , rubbery end-point
- Chapman’s reflexes at the left second intercostal space (between ribs 2 & 3) near the sternum and posteriorly between the transverse processes (TPs) T2 and T3

Somatic Findings in Hypertension

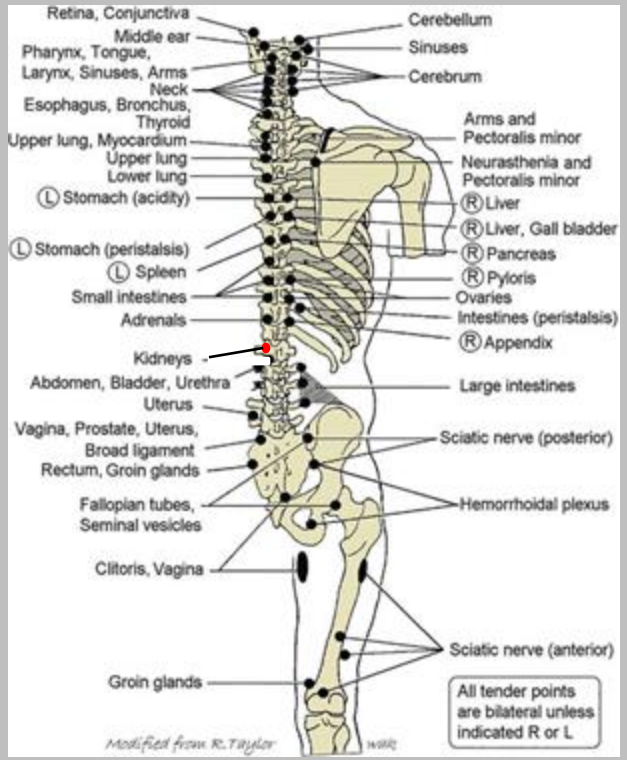
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- Adrenal Function Chapman's Point
  - Anterior: 1 inch lateral & 2 inches superior to umbilicus
  - Posterior: Intertransverse space between T11 & T12
- Kidney Chapman's points
  - Posterior: Intertransverse space between T12 & L1
  - Anterior: 1 inch superior and 1 inch lateral to the umbilicus.
- Viscerosomatic reflexes from the kidneys at T10 and T11

Anterior Chapman Points



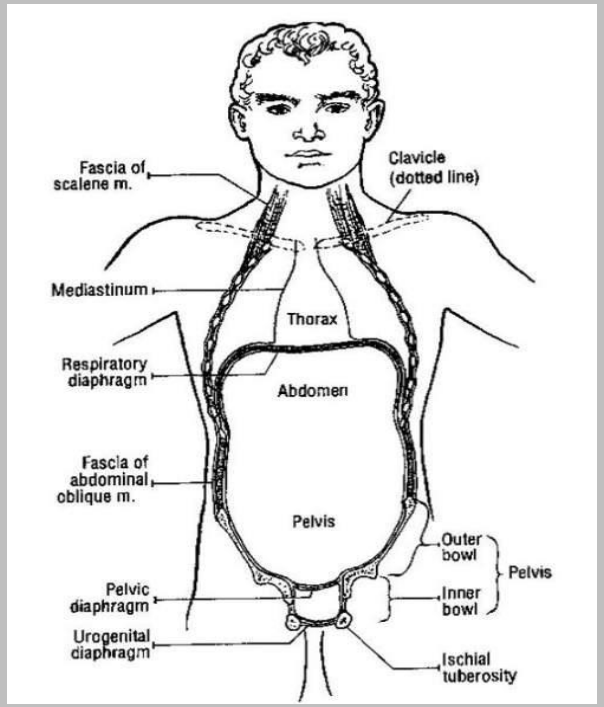
Posterior Chapmans' Points



Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018 Copyright © Wolters Kluwer

Remember there are fascial relations to the heart

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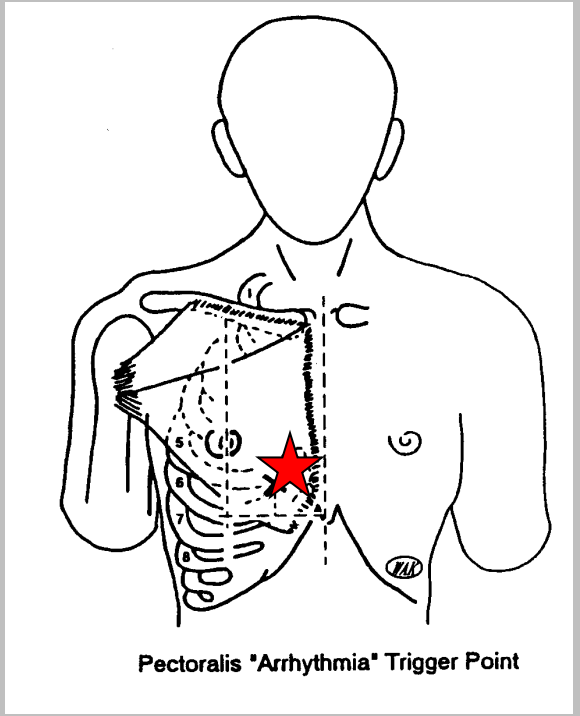


- The fibrous pericardium becomes the mediastinal fascia, and is continuous into the neck, after attaching firmly to the sternum, manubrium, and clavicles.
- In the neck it is called the pretrachial or anterior cervical fascia.
- This attaches firmly to the cranial base.

Treat fascial restrictions in the neck and thorax to optimize mediastinal motion and lymphatic flow

Pectoralis Trigger Point for Arrhythmias

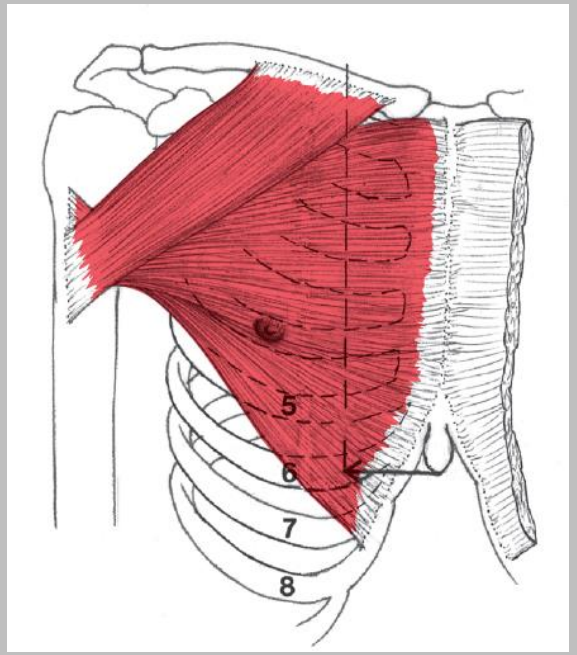
- Pectoralis Major Trigger point on right has been associated with supraventricular tachyarrhythmias.
- Located below the Right Rib 5 margin, half-way between the sternal margin & the nipple line



Kuchera. Osteopathic Considerations in Systemic Dysfunction. P 57

Travell, Simons & Amp; Simons' Myofascial Pain and Dysfunction: The Trigger Point Manual, 3e, 2019, Ch 42

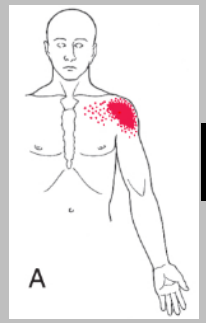
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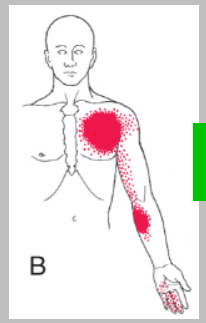
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Left sided Pectoralis Major trigger point

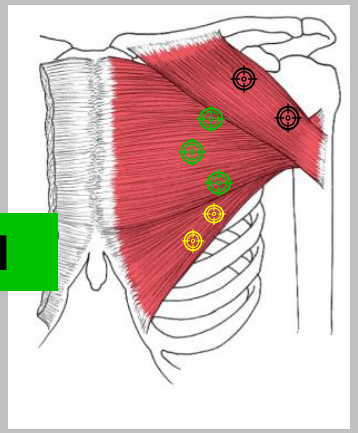
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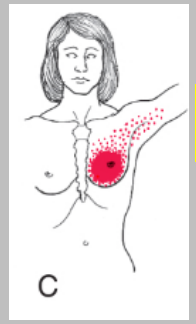


clavicular



sternal





costal

- Trigger Points of the Left Pectoralis Major May (somewhat) Resemble “Referred” Pain Pattern of Heart
- Hallmark of a Trigger Point is that the pain travels from the site of Tenderness in a characteristic pattern
- Reproducible Muscular Pain and characteristic pattern may aid clinical Dx & Tx

Travell, Simons & Amp; Simons’ Myofascial Pain and Dysfunction: The Trigger Point Manual, 3e, 2019, Ch 42

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Mitral Valve Prolapse

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- Systolic murmur with click at the **apex** of the heart.
    - Associated with a Flattened thoracic kyphosis

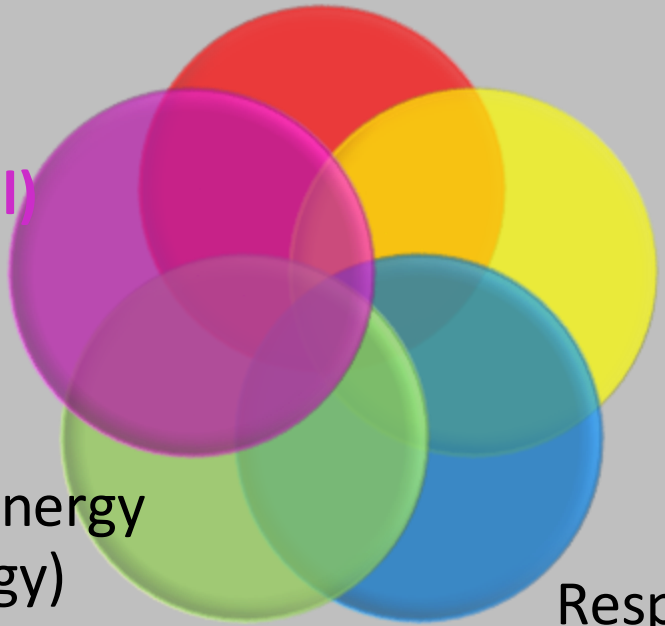
Behavioral Model

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Biomechanical
(Postural/Musculoskeletal)

Behavioral

(Psychobehavioral)



Neurological

Metabolic-Energy
(Bioenergy)

Respiratory/Circulatory

Behavioral Model

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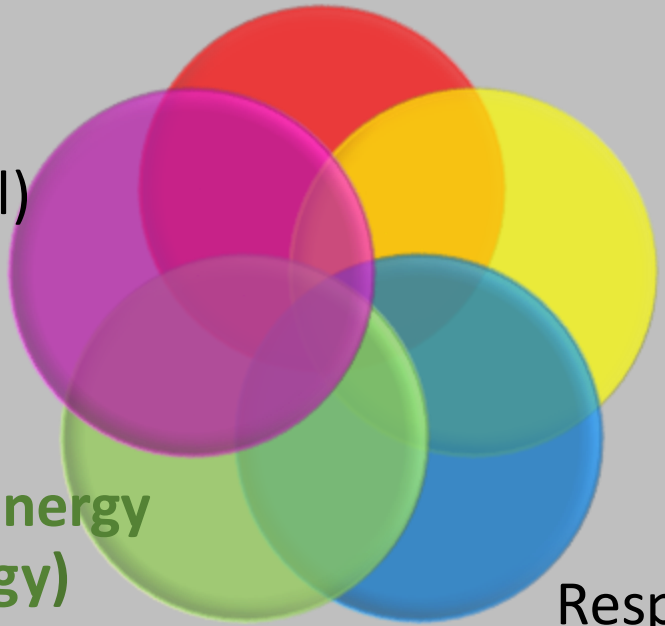
- Consider that osteopathic care (medical, manipulation, and counseling) can help
  - Decrease anxiety and stress – treating the patient allows patient to get back to work, interact with family, perform activities of daily living, etc.
  - Encourage healthy lifestyle –proper diet, exercise and sleep behaviors

Metabolic Model

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Biomechanical
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Behavioral
(Psychobehavioral)



Neurological

Metabolic-Energy
(Bioenergy)

Respiratory/Circulatory

Metabolic Model

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- One example of how OMT can affect the Metabolic-Energy model is that by treating the sympathetic nervous system will help to decrease peripheral vascular resistance, thereby decreasing the workload of the heart which will, in turn, make total body energy consumption less.

Osteopathic Tx in Cardiovascular disease

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- We will discuss the osteopathic management.
  - General medical management should always be administered first or concomitantly.

Legend: The symbol next to the model name will be used on the next 4 slides

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Biomechanical
(Postural/Musculoskeletal)

Behavioral
(Psychobehavioral)



Neurological

Metabolic-
Energy
(Bioenergy)

Respiratory/Circulatory

Treatment of Myocardial Infarction

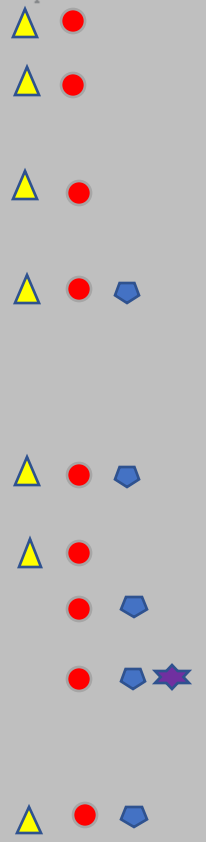
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- Treat T2-3 on left, rib raising (reducing sympathetics.)
- Post-chest compressions- ribs/sternum-indirect
- C2 & Cranial base dysfunction (vagal…parasymp)
  - Use indirect if you can
- Paraspinal inhibition to decrease the total peripheral resistance and cardiac workload (sympathetics)
- Thoracic inlet (rib 1, clavicles, scalenes): (lymphatics, stellate ganglion)
- Hyoid fascial connection to mediastinum (lymphatics)
- Pectoral traction to increase lymphatic return
- Doming of diaphragm (better motion; lymphatics)
- CV4 fluid homeostasis, decrease stress levels (parasympathetics)

Treating the Hypertensive Patient

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- T1-T5 (sympathetics to heart)
- T10-11 (sympathetics to kidneys)
- Treat All sympathetics (rib raising, chain ganglia) to decrease total peripheral resistance
- Chapman's points for the adrenal glands and kidneys.
- OMT should be slow, gentle, augment rest and relaxation, promote autonomic balance and release fascial contractures.
- OA, OM (parasympathetics)
- CV4, Fascia- Lymphatics, body fluids
- Ribs, gentle OMT- ANS, stress reduction/relaxation
- T-L junction/chapman's – ANS, balance rennin-angiotensin



Treating Arrhythmias

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- Decrease sympathetic tone
  - Cervical ganglion, rib raising T1-T5
- Modify vagal tone
  - ocular pressure, carotid massage, valsalva maneuver
  - treat occiput, OA-AA, C2
- Tx pectoral trigger point (if present)



Treating Congestive Heart Failure

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- Lymphatics- lymphatic pump, effleurage
  - Help to decrease inefficiency of the dilated thoracic duct
  - Restore lymph flow to more physiologic levels
    - Open up a path for the fluid to move, don't force the fluid back and overload an already compromised heart (ie – don't do a pedal pump)
- Treat the diaphragm and thoracic cage
  - Optimal respiratory function is vital for the maintenance of treatment effects directed at lymphatic and venous return.

Summary Slide

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- In summary, when treating any cardiovascular patient osteopathically, in addition to administering all other treatments and counseling according to the standard of care, one must address the sympathetics, parasympathetics, lymphatics and any somatic components related to the other three in order to improve homeostasis and promote physical and mental healing.
  - You are expected to be able to use the principles discussed to osteopathically evaluate and treat a patient with any cardiac disease, not only the ones mentioned.

References

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- Kuchera ML, Kuchera WA. *Osteopathic Considerations in Systemic Dysfunction Revised 2nd ed.* Columbus, OH. Greyden Press. 1994 pp53-73
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Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

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THANK YOU